

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
```

```
df = pd.read_csv('marketing_and_sales_data_evaluate_lr.csv')
df.head()
```

	TV	Radio	Social_Media	Sales
0	16.0	6.566231	2.907983	54.732757
1	13.0	9.237765	2.409567	46.677897
2	41.0	15.886446	2.913410	150.177829
3	83.0	30.020028	6.922304	298.246340
4	15.0	8.437408	1.405998	56.594181

```
df = df.dropna()
```

```
corr = df.corr()
corr['Sales'].sort_values(ascending=False)
```

	Sales
Sales	1.000000
TV	0.999497
Radio	0.868638
Social_Media	0.527446

dtype: float64

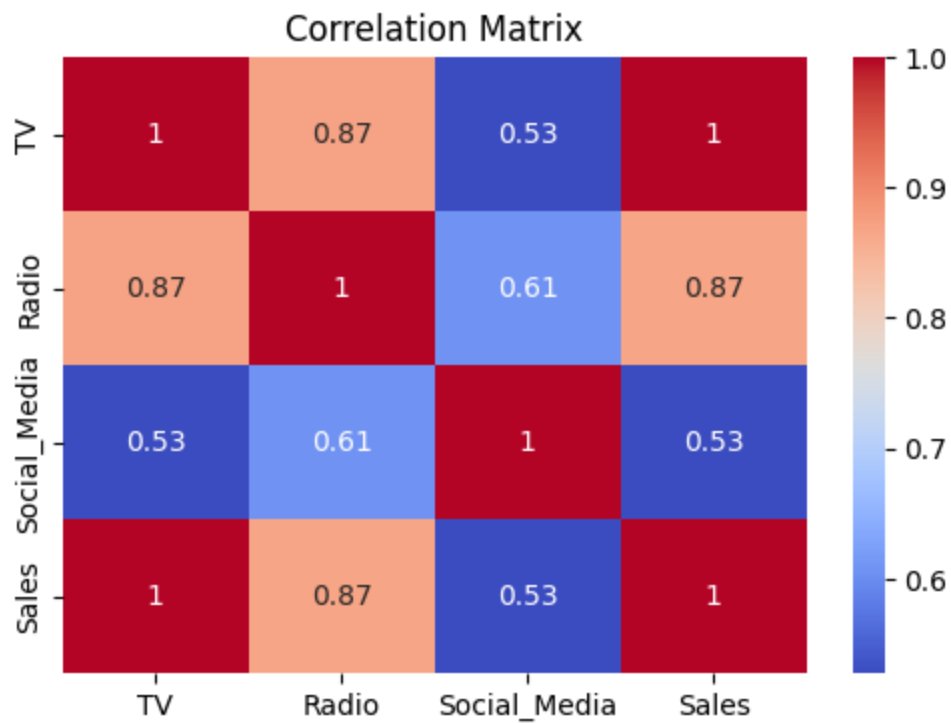
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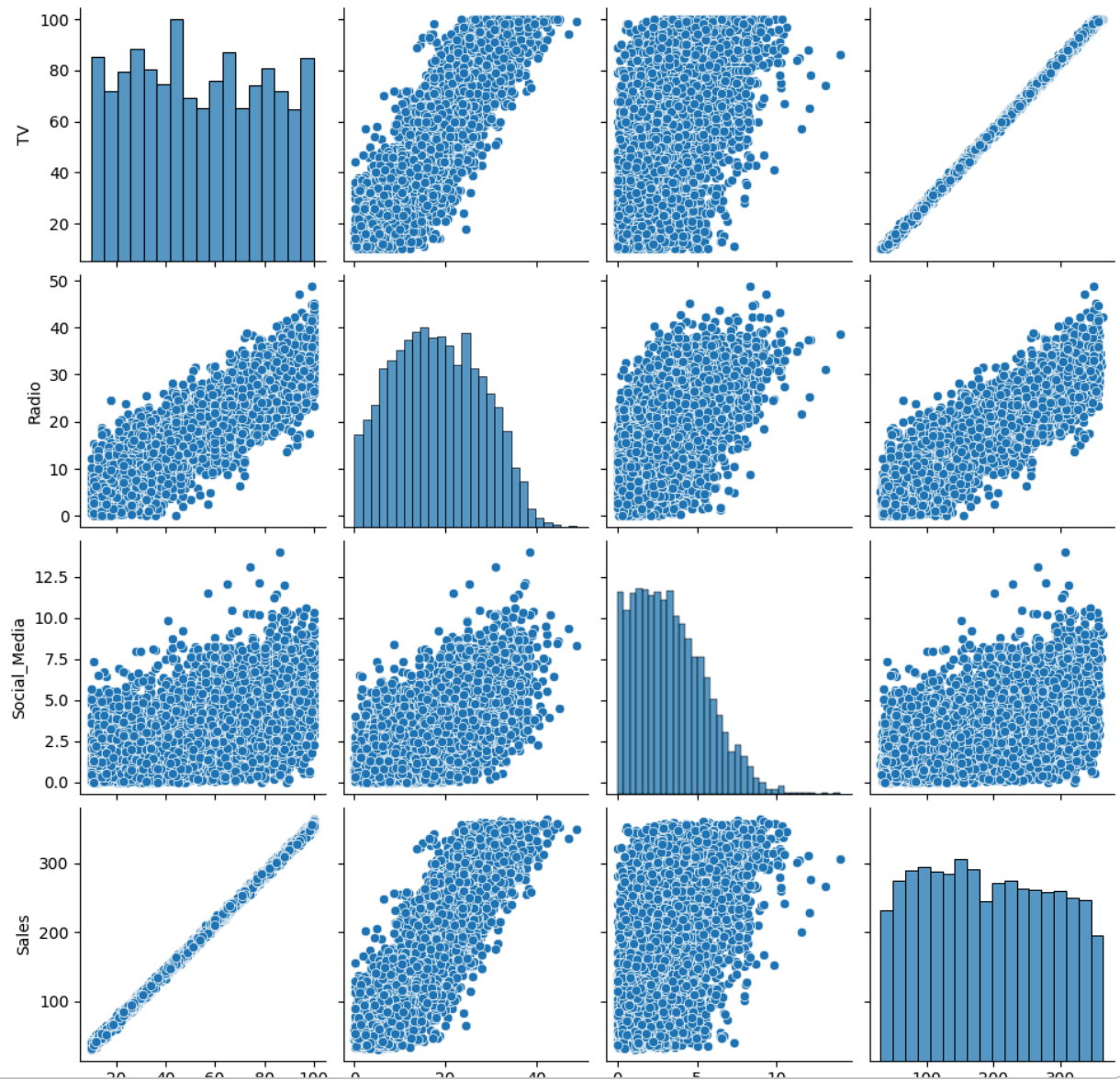
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```
plt.figure(figsize=(6,4))
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```



```
sns.pairplot(df)
plt.show()
```



```
X = df['TV']
y = df['Sales']

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()

print(model.summary())
```

OLS Regression Results

```
=====
Dep. Variable:          Sales    R-squared:
Model:                  OLS      Adj. R-squared:
Method:                 Least Squares    F-statistic:          4.51
Date:                  Mon, 08 Jun 2026    Prob (F-statistic):
Time:                  11:28:48      Log-Likelihood:        -1
No. Observations:      4546      AIC:                   2.27
Df Residuals:          4544      BIC:                   2.27
Df Model:               1
Covariance Type:       nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0
-----	-----	-----	-----	-----	-----	-----
const	-0.1325	0.101	-1.317	0.188	-0.330	
TV	3.5615	0.002	2125.272	0.000	3.558	
=====	=====	=====	=====	=====	=====	=====
Omnibus:		0.052		Durbin-Watson:		
Prob(Omnibus):		0.974		Jarque-Bera (JB):		
Skew:		-0.001		Prob(JB):		
Kurtosis:		3.012		Cond. No.		
=====	=====	=====	=====	=====	=====	=====

Notes:

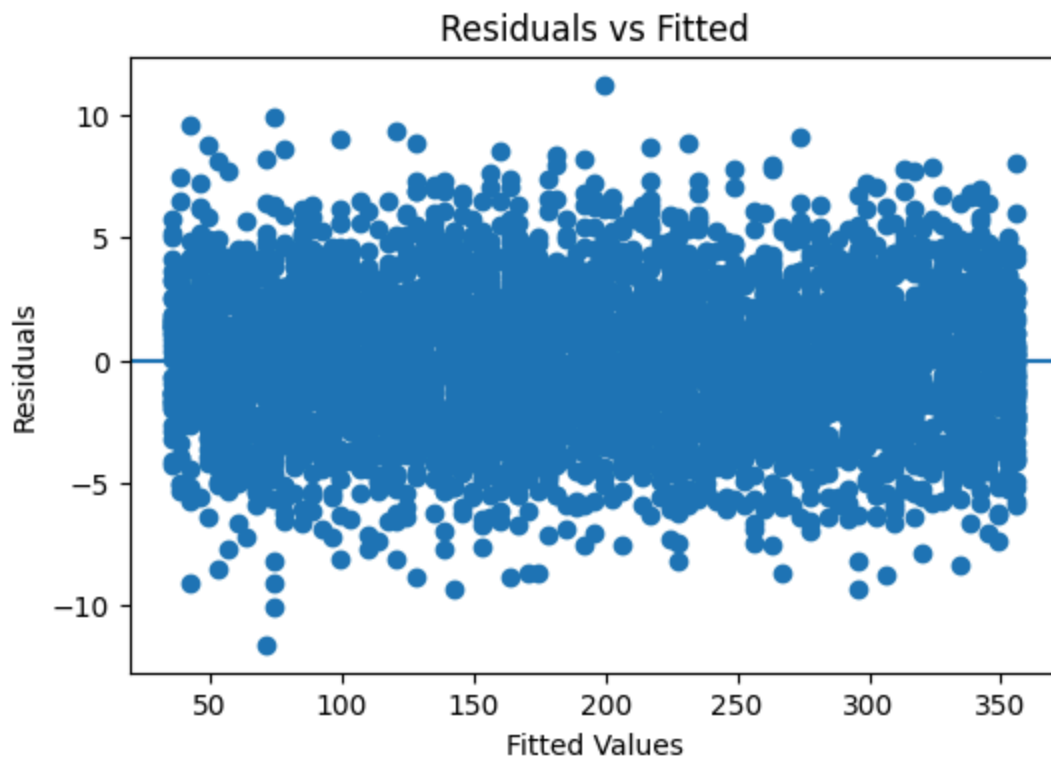
[1] Standard Errors assume that the covariance matrix of the errors is co

```

predictions = model.predict(X)
residuals = y - predictions

plt.figure(figsize=(6,4))
plt.scatter(predictions, residuals)
plt.axhline(y=0)
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
plt.title('Residuals vs Fitted')
plt.show()

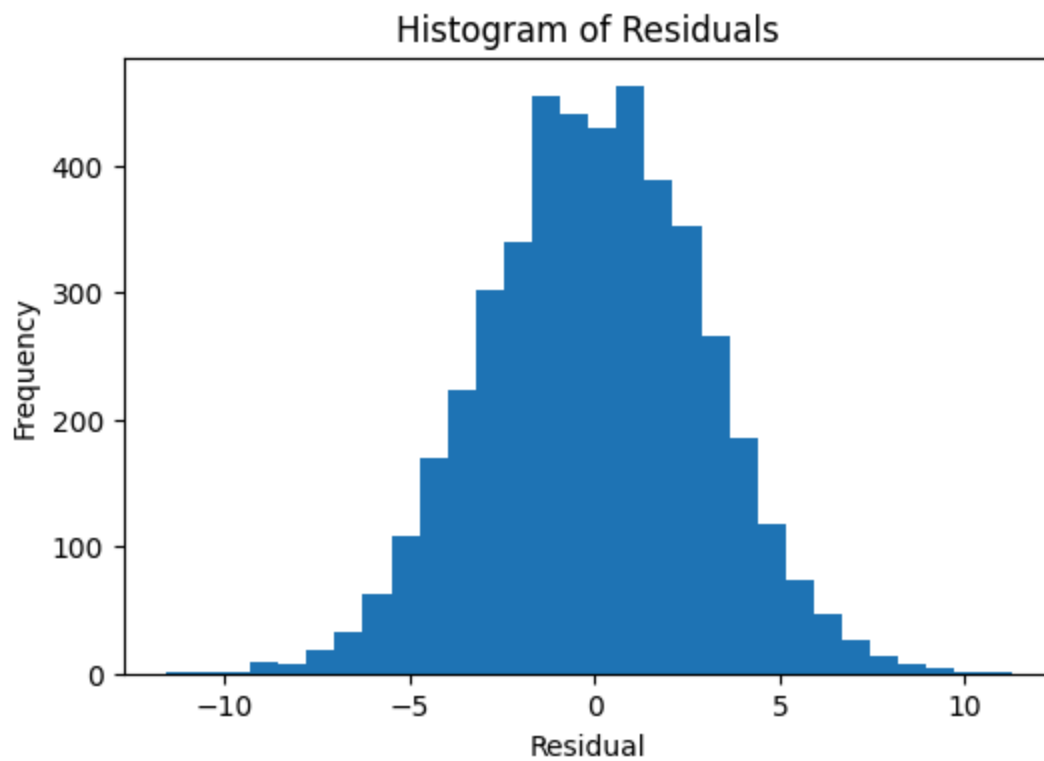
```



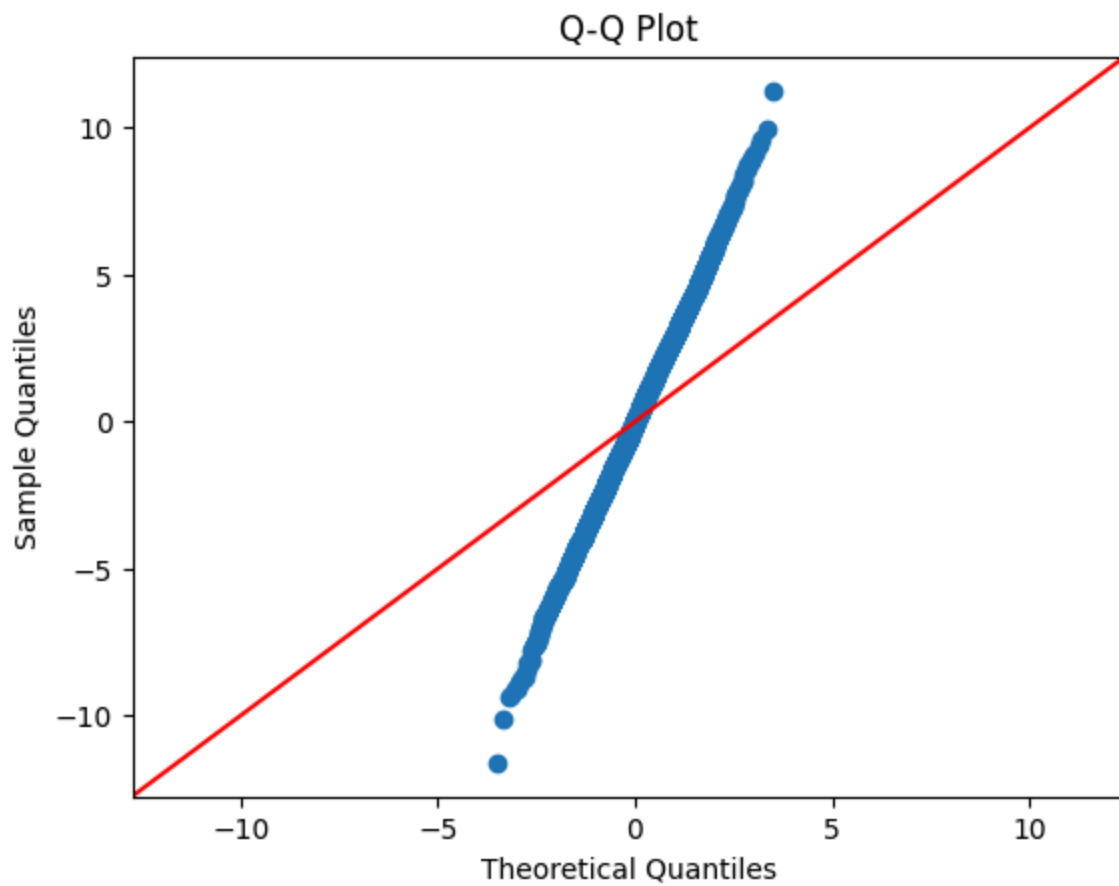
```

plt.figure(figsize=(6,4))
plt.hist(residuals, bins=30)
plt.title('Histogram of Residuals')
plt.xlabel('Residual')
plt.ylabel('Frequency')
plt.show()

```



```
sm.qqplot(residuals, line='45')  
plt.title('Q-Q Plot')  
plt.show()
```



Regression Equation

$$\text{Sales} = -0.1325 + 3.5615 \times \text{TV}$$

Interpretation: For every 1-unit increase in TV advertising expenditure, Sales increase by approximately 3.56 units.

Conclusion

TV advertising has the strongest correlation with Sales ($r = 0.9995$).

The Simple Linear Regression model achieved an R-squared value of approximately 0.999, indicating that TV advertising explains about 99.9% of the variation in Sales.

The coefficient for TV advertising is statistically significant ($p < 0.001$), indicating a strong positive relationship between TV advertising expenditure and Sales.

Business Recommendation: Organizations should prioritize investment in TV advertising because it demonstrates the strongest impact on Sales performance.